Exercise 3.2

You are given the following life table extract.

Age, x	l_x
52	89,948
53	89,089
54	88,176
55	87,208
56	86,181
57	85,093
58	83,940
59	82,719
60	81,429

Calculate

(a) ${}_{0.2}q_{52.4}$ assuming UDD (fractional age assumption),

(b) ${}_{0.2}q_{52.4}$ assuming constant force of mortality (fractional age assumption),

(c) ${}_{5.7}p_{52.4}$ assuming UDD,

(d) ${}_{5.7}p_{52.4}$ assuming constant force of mortality,

(e) ${}_{3.2|2.5}q_{52.4}$ assuming UDD, and

(f) ${}_{3.2|2.5}q_{52.4}$ assuming constant force of mortality.

Solution

(a) UDD assumption

We require the probability that a life aged 52.4 dies before reaching age 52.6. Therefore,

$${}_{0.2}q_{52.4} = 1 - {}_{0.2}p_{52.4} = 1 - \frac{l_{52.6}}{l_{52.4}}.$$

Under the uniform distribution of deaths (UDD) assumption, for $0 \le s \le 1$,

$$l_{52+s} = l_{52}(1 - sq_{52}).$$

Hence,

$$l_{52.4} = l_{52}(1 - 0.4q_{52}) \qquad \text{and} \qquad l_{52.6} = l_{52}(1 - 0.6q_{52}).$$

From the life table,

$$q_{52} = 1 - p_{52} = 1 - \frac{l_{53}}{l_{52}} = 1 - \frac{89{,}089}{89{,}948} = \frac{859}{89{,}948}.$$

It follows that

$$\begin{aligned} {}_{0.2}q_{52.4} &= 1 - \frac{1 - 0.6q_{52}}{1 - 0.4q_{52}} \\ &= \frac{0.2q_{52}}{1 - 0.4q_{52}} \\ &= \frac{0.2(859/89{,}948)}{1 - 0.4(859/89{,}948)} \\ &\approx 0.00191731656. \end{aligned}$$

Therefore,

$$\boxed{{}_{0.2}q_{52.4} \approx 1.91731656 \times 10^{-3}}.$$

(b) Constant force of mortality

Under the constant force of mortality assumption, survival over a fractional part t of the year of age is

$${}_tp_x = (p_x)^t.$$

Since the interval from age 52.4 to age 52.6 has length 0.2 and lies entirely within the year of age from 52 to 53,

$$\begin{aligned} {}_{0.2}q_{52.4} &= 1 - {}_{0.2}p_{52.4} \\ &= 1 - (p_{52})^{0.2} \\ &= 1 - \left(\frac{l_{53}}{l_{52}}\right)^{0.2} \\ &= 1 - \left(\frac{89{,}089}{89{,}948}\right)^{0.2} \\ &\approx 0.001917330671. \end{aligned}$$

Therefore,

$$\boxed{{}_{0.2}q_{52.4} \approx 1.917330671 \times 10^{-3}}.$$

(c) UDD assumption

Survival for 5.7 years from age 52.4 means survival to age

$$52.4 + 5.7 = 58.1.$$

Therefore,

$${}_{5.7}p_{52.4} = \frac{l_{58.1}}{l_{52.4}}.$$

Under the UDD assumption, the number of survivors is interpolated linearly within each year of age. Thus,

$$l_{52.4} = l_{52} - 0.4d_{52} \qquad \text{and} \qquad l_{58.1} = l_{58} - 0.1d_{58},$$

where

$$d_{52} = l_{52} - l_{53} = 89{,}948 - 89{,}089 = 859$$

and

$$d_{58} = l_{58} - l_{59} = 83{,}940 - 82{,}719 = 1{,}221.$$

Hence,

$$\begin{aligned} {}_{5.7}p_{52.4} &= \frac{l_{58} - 0.1d_{58}}{l_{52} - 0.4d_{52}} \\ &= \frac{83{,}940 - 0.1(1{,}221)}{89{,}948 - 0.4(859)} \\ &= \frac{83{,}817.9}{89{,}604.4} \\ &\approx 0.935421698. \end{aligned}$$

Therefore,

$$\boxed{{}_{5.7}p_{52.4} \approx 0.935421698}.$$

(d) Constant force of mortality

As in part (c), survival for 5.7 years from age 52.4 is survival to age 58.1, so

$$ {}_{5.7}p_{52.4} = \frac{l_{58.1}}{l_{52.4}}.$$

Under the constant force of mortality assumption, fractional-year survival is exponential. Therefore,

$$l_{52.4} = l_{52}(p_{52})^{0.4} \qquad \text{and} \qquad l_{58.1} = l_{58}(p_{58})^{0.1}.$$

Using

$$p_{52} = \frac{l_{53}}{l_{52}} \qquad \text{and} \qquad p_{58} = \frac{l_{59}}{l_{58}},$$

we obtain

$$\begin{aligned} {}_{5.7}p_{52.4} &= \frac{l_{58}(p_{58})^{0.1}}{l_{52}(p_{52})^{0.4}} \\ &= \frac{83{,}940\left(\dfrac{82{,}719}{83{,}940}\right)^{0.1}}{89{,}948\left(\dfrac{89{,}089}{89{,}948}\right)^{0.4}} \\ &\approx 0.935423024785. \end{aligned}$$

Therefore,

$$\boxed{{}_{5.7}p_{52.4} \approx 0.935423025}.$$

(e) UDD assumption

The deferred mortality probability ${}_{3.2|2.5}q_{52.4}$ is the probability that a life aged 52.4 survives for 3.2 years and then dies during the following 2.5 years. Therefore,

$$\begin{aligned}
{}_{3.2|2.5}q_{52.4} &= {}_{3.2}p_{52.4}\,{}_{2.5}q_{55.6} \\
&= \frac{l_{55.6}}{l_{52.4}}\left(1-\frac{l_{58.1}}{l_{55.6}}\right) \\
&= \frac{l_{55.6}-l_{58.1}}{l_{52.4}}.
\end{aligned}$$

Under UDD, the required fractional-age survivor values are

$$\begin{aligned}
l_{52.4} &= l_{52} - 0.4d_{52} = 89{,}948 - 0.4(859) = 89{,}604.4, \\
l_{55.6} &= l_{55} - 0.6d_{55} = 87{,}208 - 0.6(87{,}208 - 86{,}181) = 86{,}591.8, \\
l_{58.1} &= l_{58} - 0.1d_{58} = 83{,}940 - 0.1(1{,}221) = 83{,}817.9.
\end{aligned}$$

Hence,

$$\begin{aligned}
{}_{3.2|2.5}q_{52.4} &= \frac{86{,}591.8 - 83{,}817.9}{89{,}604.4} \\
&\approx 0.030957185138.
\end{aligned}$$

Therefore,

$$\boxed{{}_{3.2|2.5}q_{52.4} \approx 0.030957185}.$$

(f) Constant force of mortality

As in part (e), the deferred mortality probability can be written as

$$\begin{aligned}
{}_{3.2|2.5}q_{52.4} &= {}_{3.2}p_{52.4}\,{}_{2.5}q_{55.6} \\
&= {}_{3.2}p_{52.4}\left(1 - {}_{2.5}p_{55.6}\right) \\
&= {}_{3.2}p_{52.4} - {}_{5.7}p_{52.4} \\
&= \frac{l_{55.6}-l_{58.1}}{l_{52.4}}.
\end{aligned}$$

Under the constant force of mortality assumption,

$$\begin{aligned}
l_{52.4} &= l_{52}(p_{52})^{0.4}, \\
l_{55.6} &= l_{55}(p_{55})^{0.6}, \\
l_{58.1} &= l_{58}(p_{58})^{0.1}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
{}_{3.2|2.5}q_{52.4} &= \frac{l_{55}(p_{55})^{0.6}}{l_{52}(p_{52})^{0.4}} - \frac{l_{58}(p_{58})^{0.1}}{l_{52}(p_{52})^{0.4}} \\
&= \frac{87{,}208\left(\dfrac{86{,}181}{87{,}208}\right)^{0.6} - 83{,}940\left(\dfrac{82{,}719}{83{,}940}\right)^{0.1}}{89{,}948\left(\dfrac{89{,}089}{89{,}948}\right)^{0.4}} \\
&\approx 0.030950242840.
\end{aligned}$$

Therefore,

$$\boxed{{}_{3.2|2.5}q_{52.4} \approx 0.030950243}.$$